International Journal of Tomography & Statistics,
Fall 2010, Vol. 15, No. F10; Int. J. Tomogr. Stat.,
ISSN 0972-9976 (Print); ISSN 0973-7294 (Online)
Copyright © 2010 by IJTS (CESER Publications)

Multichannel image restoration using combined channel information and robust M-estimator approach

V. B. Surya Prasath and Arindama Singh

Department of Mathematics
Indian Institute of Technology Madras
Chennai, 600036, India
surya.iit@gmail.com asingh@iitm.ac.in

ABSTRACT

We propose a novel multichannel image restoration scheme based on Bayesian maximum a posteriori estimation. Restoration schemes for mono-channel images fail in the case of multichannel images since they do not take into account the cross channel relations present. We propose a robust M-estimator based Gibbs prior which uses not only spatial but also interchannel term for multichannel image data modelled as a Markov random field. Regularized total variation potential function is used as a robust M-estimator in each channel and a weighted inter-channel coupling term with Gaussian prior is used to capture the cross-correlations of the given image. We prove the well-posedness of the corresponding minimization in the bounded variation space. This variational Bayesian scheme reduces the channel mixup artifacts and preserves color edges when compared with other schemes on real and noisy multidimensional images.

Keywords: Markov Random Fields, Robust M-estimators, Variational Bayesian approach, Energy Minimization, Multichannel Image Restoration.

2000 Mathematics Subject Classification: 62M40, 49J45, 68U10, 94A08.

1 Introduction

Multichannel images, in general, and color images, in particular, are now prevalent in scientific applications. Charged-coupled device sensor based cameras output color image data with three primary color channels red, green and blue, whereas satellite data consists of a range of frequencies and usually outputs multi-spectral image, for example in remote sensing applications. However due to limitations in the imaging system and imaging environments the output image often is noisy and blurred. Hence there is a need to post-process the data to get an estimate of the latent data. Monochromatic image restoration (Aubert and Kornprobst, 2002) is well studied by many researchers. A simple solution is to apply these schemes to each channel image and then combine the output into a multichannel one. However if cross channel relation is present then these methods cannot give high quality results.

Starting with linear models based on Gaussian priors such as Wiener filtering, least squares restoration, and various other models using the Bayesian approach (Geman and Geman, 1984;

Li, 1995; Li, 1998) have been introduced for the gray scale images. Using Gibbs priors and statistical estimators Markov random field (MRF) modelling (Li, 1995; Ceccarelli, 2007) is well suited for images and multidimensional data; see (Szeliski et al., 2008) for a recent review on energy minimization methods for MRFs. The a priori term incorporates the smoothness assumptions about the unknown image and convex robust M-estimators give rise to well posed minimization schemes. Also nonconvex potentials (Black et al., 1998; Szeliski et al., 2008) have been proposed despite the theoretical problems associated with them. Due to the advent of computers and the availability of commercial digital cameras multichromatic image processing is an exciting field and an active research area. Extension of the previous linear and nonlinear models to multichannel case has been done successfully, although capturing the cross channel relations is an open area of research.

In this paper we propose a novel and robust multichannel image model for use in a non-Gaussian Markov random field using the Bayesian paradigm. An M-estimator based edge preserving potential function is used in intrachannel restoration whereas a weighted coupling function is used to integrate inter-channel correlations in the digital image. This gives a well-posed convex minimization problem via gradient descent and is computationally tractable. We also make use of nonconvex Lorentzian M-estimator function and present its results. Advantage of incorporating cross correlation between all the channels using stochastic modelling is established on various real and noisy images.

The rest of the paper is organized as follows. In Section 2 we study the MRF modelling for multichannel images via variational Bayesian approach and in Section 3 we propose our improved scheme using M-estimator based prior and a coupling term. In Section 4 a detailed comparison of the proposed approach with channel-wise and other schemes with numerical simulations are shown. Section 5 concludes the paper.

2 Multichannel Gibbs-MRF model

In this section we review a general scheme based on Maximum a posteriori Markov random field (MAP-MRF) formulation for multichannel image denoising problem. Assuming the true data $u : \Omega \to \mathbb{R}^N$ is corrupted by noise, the noisy input N-channel image I is given by:

$$I = u + n \tag{2.1}$$

where $\Omega \subset \mathbb{R}^2$ (usually a rectangle) and n denotes the random noise present in the system (assumed to be iid, identically and independently distributed). We also assume that the noise n is uncorrelated to the latent data u. To solve the problem (2.1) in Bayesian paradigm we consider the estimate $\hat{u}$ with the maximum *a posteriori* probability given I:

$$\hat{u} = arg \max_u Pr(u|I) \tag{2.2}$$

The conditional probability is given by the Bayes Theorem

$$Pr(u|I) := \frac{Pr(I|u)Pr(u)}{Pr(I)}$$

It is a product between the *likelihood* $Pr(I|u)$, which depends on the noise distribution and the *prior* $P(u)$, which covers our *a priori* assumption about the true image. We assume that the likelihood to be Gaussian density distribution of standard deviation σ, and since it is uncorrelated to u, $Pr(I|u)$ is a probability model for the noise corrupting the data, i.e,

$$Pr(I|u) = Pr(I-u|u) = Pr(n|u) = Pr(n) = \exp(-\|u-I\|_2^2/2\sigma^2)$$

Note that this additive noise models white noise, for example thermal noise in Charge-coupled device (CCD) acquisition sensors and standard camera noise. Now by the following theorem (Li, 1995), a priori smoothness assumption on u given by $Pr(u)$ follows a Gibbs distribution.

Theorem 2.1 (Clifford-Hammersley Theorem)**.** *Let the u be the random field defined on the set of m points in $\mathbb{R}$. Then u is an MRF if only if the distribution of u has the form of a Gibbs distribution*

$$Pr(u) := \exp\left(-\sum_{c\in\mathcal{N}} V_c(u)\right) = \exp\left(-E(u)\right),$$

where $V_c(\cdot)$ is any function of a local group of points c, and $\mathcal{N}$ is the set of all such local groups.

Thus the overall conditional probability (2.2) becomes

$$Pr(u|I) = \frac{\exp\left(-E(u|I)\right)}{Pr(I)} = \frac{\exp\left(-[\|u-I\|_2^2 + E(u)]\right)}{Z},$$

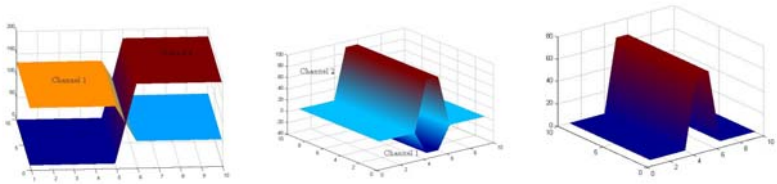
where Z is a normalization constant. $E(u)$ is called the energy functional and it measures the quality of the image data u. $E(u)$ is the sum of local contributions from each pixel; therefore, it should be aimed at measuring the discontinuities of the solution u. Such discontinuities depend on the magnitude of the derivative of u. Classical least square (LS) method uses $E(u) = \|\nabla u\|_2^2 := \int |\nabla u|^2$, where ∇u is the gradient computed locally around a pixel i using neighborhood pixels $j \in \mathcal{N}_i$; for example the discrete gradient at j can obtained by $\nabla u_j = u_j - u_i$. The clique $\mathcal{N}_i$ defines a local group of pixels around i. The main drawback in LS method is that it disallows discontinuities in the solution. To remedy this we can use an edge preserving scheme with a prior term of the form

$$E(u) = \sum_{i=1}^{N} E(u^i) = \sum_{i=1}^{N} \int_\Omega \phi(\nabla u^i)\, dx \tag{2.3}$$

where ϕ is called the *potential function* in robust statistics (Huber, 1981). The idea is to reduce the growth of ϕ for large differences $|u_j - u_i|$, which corresponds to jumps or discontinuities in the data. In this paper we make use of such potentials (Hampel et al., 1986) from robust statistics.

By minimizing the negative log of the probability (2.2) we get the solution to Bayesian MAP problem as

$$\hat{u} = arg\min_u E(u|I), \tag{2.4}$$



(a) (b) (c)

Figure 1: Cross channel edge case and the effect of the coupling weight term: (a) Two channel data (u^1, u^2) with a step edge of different magnitude (b) $(\nabla u^1, \nabla u^2)$ (c) Same value in the weight terms $\omega^2 \nabla u^1$ and $\omega^1 \nabla u^2$ using quadratic coupling term.

where the energy is of the form

$$E(u|I) = \int_\Omega (u - I)^2 \, dx + \lambda \sum_{i=1}^{N} \int_\Omega \phi(\nabla u^i) \, dx \tag{2.5}$$

The parameter $\lambda > 0$ controls the influence of smoothing strength function. It provides a tradeoff between prior assumptions and a fit to the observed data.
The discontinuity adaptive function ϕ is chosen to be an even function in a way that there exists a function g with $\phi'(s) = 2sg(s)$. The adaptive interaction function (AIF) g decides the smoothing and outlier rejection (Li, 1998). Note that under Gaussian noise assumption the likelihood term (also called the data fidelity term) is quadratic and hence convex in u. Thus if the prior term is also convex in u then we are guaranteed of well-posedness of the energy minimization scheme (2.4-2.5).

3 Coupled intra channel term

The energy term $E(u)$ in (2.5) does not take into account the multichannel nature and ignores the cross channel information present in digital images. For example, Figure 1 shows an instance where two channels have different edge magnitudes at a step edge. In this case the second channel u^2 has larger magnitude ∇u^2 than the first channel u^1 of magnitude ∇u^1. If one channel edge is bigger (gradient magnitude larger) than the other channel edge then it should be reflected in the restoration process and preserved by it. Using this reasoning we propose to add an intra-channel coupling term of the following form to the prior term (2.3):

$$E(u^i, u^j) = \int_\Omega \psi(\omega^i(\nabla u)\nabla u^j - \omega^j(\nabla u)\nabla u^i) \, dx \tag{3.1}$$

The weights ω^i use the gradient at channel i and the function ψ penalizes large differences via the energy minimization (2.4). One possible choice is to use the smoothed input gradients with a Gaussian kernel $G_\sigma(x) = (2\pi\sigma^2)^{-1} \exp(-x^2/2\sigma^2)$. In this case,

$$\omega^i(\nabla u) = \nabla I^i_\sigma, \tag{3.2}$$

where $I^i_\sigma = G_\sigma \star I^i$ and $\star$ is the convolution operation. Note that smoothing is required to remove noise outliers in computing gradients from the input image I. For penalizing the misalignment

of edges between channels, we use an LS function $\psi(s) = s^2$ as a prior about inter-channel coupling effect. Thus we get the minimization problem for a N-dimensional image as

$$E(u|I) = \int_\Omega (u-I)^2\,dx + \lambda \sum_{i=1}^{N} \int_\Omega \phi(\nabla u^i)\,dx + \mu \sum_{i=1}^{N}\sum_{j=1}^{N} \int_\Omega (\nabla I_\sigma^i \nabla u^j - \nabla I_\sigma^j \nabla u^i)^2\,dx, \quad (3.3)$$

where $\mu > 0$ controls the influence of the coupling term. Next, we prove that in the case of convex prior functions the proposed scheme (3.3) is well posed in bounded variation space. For the preliminaries and results about bounded variation space $BV(\Omega)$ we refer the reader to (Giusti, 1984).

Theorem 3.1 (Well-posedness)**.** *Let $I \in BV(\Omega;\mathbb{R}^N)$. Suppose the function $\phi : \mathbb{R} \to \mathbb{R}^+$ is convex and satisfies the following conditions:*

1. *ϕ is an even function, nondecreasing in $\mathbb{R}^+$ with $\phi(0) = 0$,*
2. *there exist $c > 0, b \geq 0$ such that $cz - b \leq \phi(z) \leq cz + b$, $\forall z \in \mathbb{R}^+$.*

Then the minimization problem (2.4) with the energy $E(u|I)$ given by (3.3) has a solution $\hat{u} \in BV(\Omega;\mathbb{R}^N)$ and the solution is unique if ϕ is strictly convex.

Proof. Let us write the total coupling term as

$$E_c(u) = \sum_{i=1}^{N}\sum_{j=1}^{N} \int_\Omega (\nabla I_\sigma^i \nabla u^j - \nabla I_\sigma^j \nabla u^i)^2\,dx$$

Note that the data fidelity term and $E_c(u)$ are quadratic and hence strictly convex. Extend the functional $E(u|I)$ in (2.5) to $BV(\Omega;\mathbb{R}^N)$ with weak*-topology by an application of Theorem 3.2.1 in (Aubert and Kornprobst, 2002) to get the relaxed functional

$$\tilde{E}(u|I) = \int_\Omega (u-I)^2\,dx + \sum_{i=1}^{N} \int_\Omega \phi(\nabla u^i)\,dx + c\left(\sum_{i=1}^{N} \int_{\Omega\setminus S} \left|\nabla_s u^i\right|\right),$$

where $c = \lim_{s\to+\infty} \frac{\phi(s)}{s}$ and S denotes the jump set and $\nabla_s u$ is the singular part of the gradient. Then the use of conditions on ϕ implies (see for example Theorem 3.2.2 in (Aubert and Kornprobst, 2002)) that the extended functional has a minimizer with

$$\min_u \tilde{E}(u|I) = \inf_{u\in BV(\Omega;\mathbb{R}^n)} E(u|I),$$

and it is unique if ϕ is strictly convex. □

For the interchannel smoothing term we use the regularized L^1 norm potential function, satisfying the conditions of Theorem 3.1 in the form:

$$\phi(\nabla u^i) = \sqrt{(\nabla u^i)^2 + \epsilon^2} \quad (3.4)$$

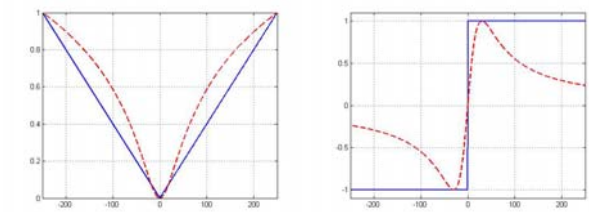


Figure 2: Robust M-estimators as potential functions ϕ (left) and their corresponding AIF functions g (right): (solid line) RTV (L^1 norm) (3.4) and (dashed line) Lorentzian (3.5).

This function is also known as regularized total variation (RTV) (Rudin et al., 1992). Other choices are also possible (Black et al., 1998), for example, the Lorentzian function

$$\phi(\nabla u^i) = \log\left[\frac{k^2 + (\nabla u^i)^2}{k^2}\right] \tag{3.5}$$

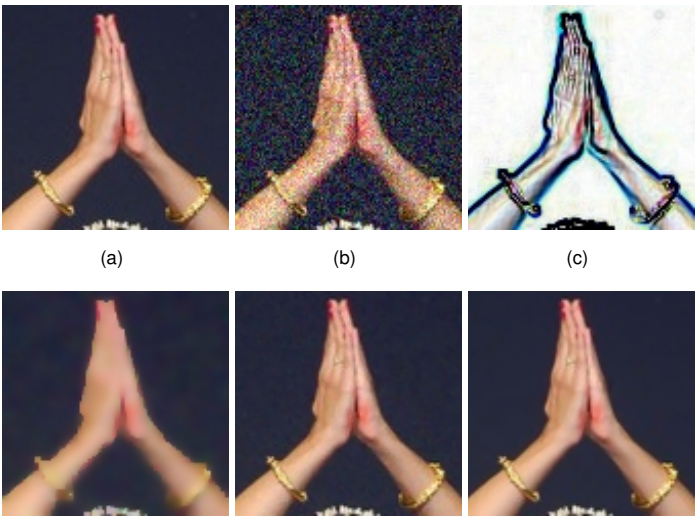
which is nonconvex and the outliers are penalized linearly. Figure 2 shows both the prior potential functions and their AIF functions. Although the Lorentzian function (3.5) is not convex and hence the above existence result (Theorem 3.1) does not apply, still it gives numerically stable denoising results; see Section 4. The above theorem can be in general proved for the general coupling term (3.1) under some assumptions on the inter-channel prior function $\psi : \mathbb{R} \to \mathbb{R}^+$ and will be reported elsewhere. The advantages of the proposed scheme (3.3) with robust M-estimators ϕ satisfying the conditions of Theorem 3.1 are as follows:

- Selective smoothing and outlier removal due to the term (2.3) in each image planes.
- All channel edges are aligned and large deviations are penalized under the term (3.1) giving the output image a coherent look.
- At places where one channel has an edge and the other a homogenous region the weight term (3.2) compensates it by using the initial gradient estimates.

4 Numerical Results and discussion

The proposed MAP-MRF multichannel formulation (3.3) with the regularized L^1 norm potential function (3.4) gives a well-posed scheme due to Theorem 3.1. To implement the scheme we use gradient descent with empirically chosen parameters λ and μ and the presmoothing Gaussian width in the coupling term (3.2) is fixed at $\sigma = 1$ for the reported images. To implement the channel-wise scheme we took $\mu = 0$. The regularization of the L^1 norm as a prior term for each image plane is done with $\epsilon = 10^{-6}$ in all the experiments and for Lorentzian M-estimator we set the threshold parameter $k = 30$.

Figure 3(a) shows 256×256 *Hands* color image, taken by a commercial digital camera; it contains unknown shot noise and less visible jpeg artifacts. Further we add additive Gaussian noise, Fig. 3(b), $\sigma = 20$ on all the three channels (Red, Green, Blue) testing the proposed



(a) (b) (c)

(d) (e) (f)

Figure 3: Hands RGB image denoising: (a) original color image (b) noisy image (c) combined edge information in all the three channels (d) channel-wise RTV scheme $\mu = 0, \epsilon = 10^{-6}$ (e) coupled Lorentzian scheme $\lambda = 30, \mu = 20, k = 30$ (f) coupled RTV scheme $\lambda = 30, \mu = 30, \epsilon = 10^{-6}$.

scheme (3.3). It can be seen by comparing Fig. 3(d-f) that the proposed coupling schemes with L^1 term (3.4) and with Lorentzian function (3.5) performs better than the channel-wise method. Channel RTV scheme outputs a piecewise constant image (Fig. 3(d)) and color smearing happens near prominent edges. Due to the coupling term edges in all the three color planes (Fig. 3(c)) are taken into account giving a much better result in terms denoising and edge preservation. Coupled RTV (Fig. 3(f)) gives a sharper result than coupled Lorentzian (Fig. 3(e)) due to the application of total variation (Rudin et al., 1992) in interchannel computations. In channel-wise restoration (Fig. 3(d)) colors are mixed and the center edge in the hand is not recovered well. In coupled RTV and coupled Lorentzian schemes there is no color smearing due to the term (3.1) and edges are well preserved.

Comparison with two other relevant schemes were undertaken to show the effectiveness of our model. Brook, Kimmel and Sochen (BKS) (Brook et al., 2003), propose to use the interchannel term $|\nabla_x u \times \nabla_y u|^2$, where the derivatives $(\nabla_x u, \nabla_y u)$ are computed in their corresponding (x, y) axis directions. This helps in aligning the color edges as in our model. Moreover the intrachannel term is chosen to be $\phi(\nabla u^i) = (\nabla u^i)^2$. The corresponding wellposedness of the minimization problem and its numerical convergence are not proven. Also, the term $|\nabla_x u \times \nabla_y u|^2$ is of order $|\nabla u|^4$ and cannot be bounded below by $c\,|\nabla u|^4$ for some $c > 0$, thus we cannot prove a similar result such as Theorem 3.1 in our work.

Karen and Goltlib (KG) (Karen and Gotlib, 1998) propose to use the discrete interchannel term

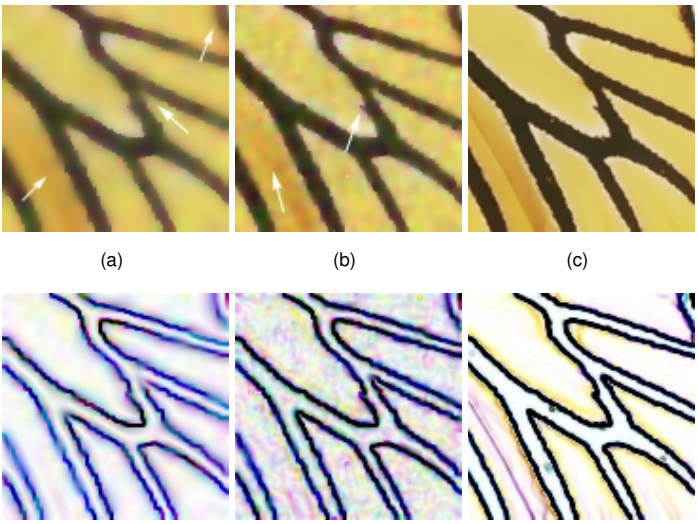


Figure 4: Comparison with other schemes: (a) KG scheme (b) BKS scheme (c) proposed scheme (3.3-3.4) $\lambda = 20, \mu = 50, \epsilon = 10^{-6}$. The arrows show areas where the marked improvement in our scheme is observed and the other two schemes fail.

$E_c(u) = \sum_\Omega \sum_{N_{ij}} |P_{ij} \times P_{kl}|^2$, where P_{ij} denotes the vector of (R, G, B) values at a pixel (i, j) and the cross product is taken in a 3×3 neighborhood N_{ij} of the pixel P_{ij}. Moreover the intrachannel term is chosen to be a second order term $\phi(\nabla u^i) = (u^i_{xx})^2 + 2(u^i_{xy})^2 + (u^i_{yy})^2$. Due to the isotropic smoothing nature of this interchannel term, the edges of each channel is smoothed and the localization of the combined edges are reduced. Also, the scheme is nonquadratic and the product term is biased toward areas with higher color values. In Fig. 4 restoration of the *Monarch* image corrupted by noise $\sigma = 30$ is compared with these two schemes. In KG scheme output (Fig. 4(a)) oversmoothing occurs near corners and loses fine scale edges (Fig. 4(d), whereas in BKS scheme output (Fig. 4(b)) due to the noise prone cross product gradient term oscillations occur (Fig. 4(e)). Compare this with our scheme output image in Fig. 4(c) and its edge map Fig. 4(f) which retains all the multiscale edges, corners (see for example, the left hand corner where a thin edge is preserved by our scheme).

Figure 5 shows the denoising of a multispectral data of the Mississippi River acquired by the Spaceborne Imaging Radar-C/X-band Synthetic Aperture (SIR-C/X-SAR) system. The different colors represent the data return in different radar channels: red is L-band, vertically transmitted and received; green is L-band vertically transmitted and horizontally received; and blue is C-band vertically transmitted and received. The restoration using L^1 norm for each of the corresponding channels is shown in Fig. 5(d-f). The outliers correspond to noise are removed and the edges in each image plane is preserved. The combined image (Fig. 5(h)) shows better edge coherency and the river boundary and farmlands with variety of colors representing crops surrounding it are also preserved.

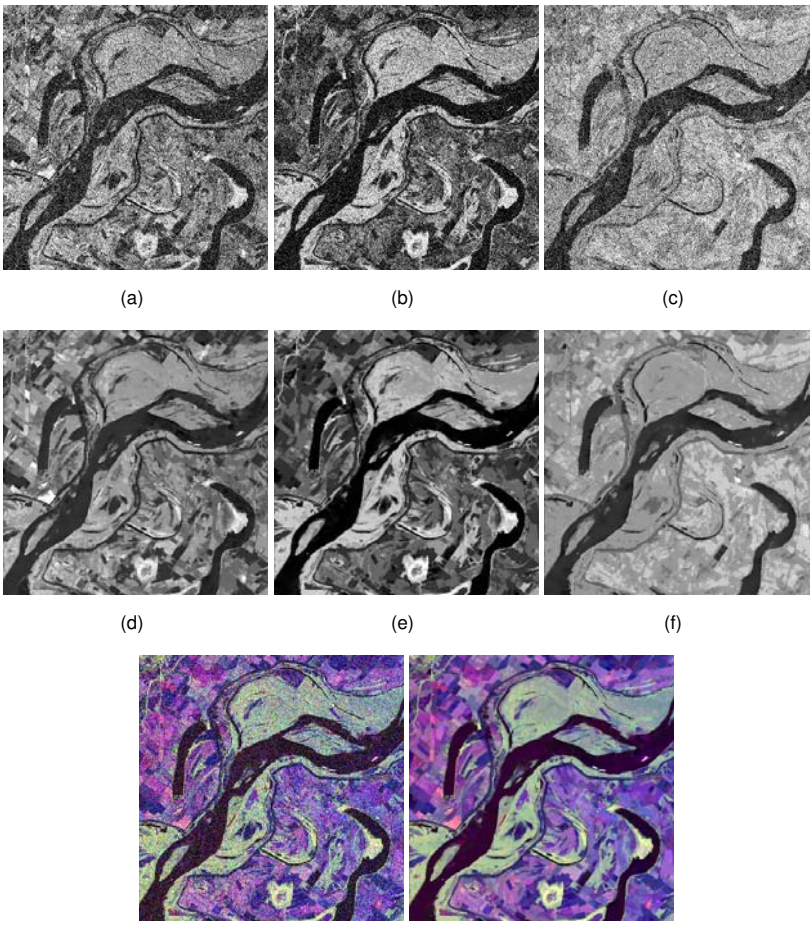


Figure 5: Multispectral image denoising: (a-c) L-band (VTVR), L-band (VTHR) and C-band (VTVR) (d-f) corresponding denoised bands (g) combined image (h) denoised using the proposed coupled L^1 norm scheme (3.3-3.4).

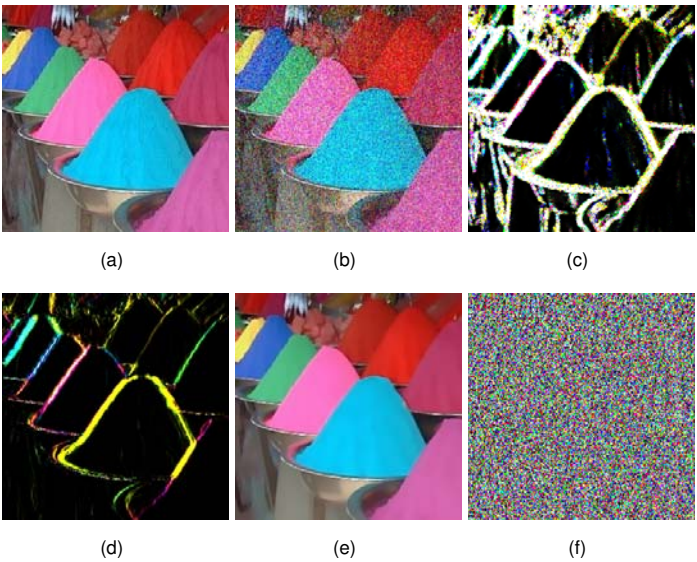


Figure 6: High noise removal: (a) original Colors 280×280 image (b) gaussian noise added ($\sigma = 40$) (c) the adaptive coupling term (d) unit weights coupling term (e) denoised using the adaptively updated coupled term with L^1 norm M-estimator $\lambda = 30, \mu = 50, \epsilon = 10^{-6}$ (f) Residual image.

Next example shows a high noise case in Fig. 6(b). In this experiment we update the weight term (3.1) adaptively in the following manner instead of the input image gradients (3.2)

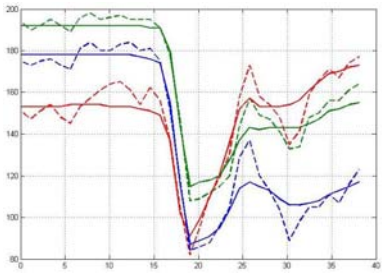
$$\omega^i(\nabla u) = \nabla u^i_\sigma$$

This intra channel distribution $(E(u^1, u^2), E(u^1, u^3), E(u^2, u^3))$ is shown in Fig. 6(c) in (R, G, B) channels. We also show the the unit weight distribution, i.e, $\omega^i \equiv 1$, clearly adaptive coupling term has more structural information and captures multiscale nature of the digital image. The resulting image obtained with L^1 norm M-estimator (Fig. 6(e)) shows the marked improvement and there are no color smearing between different color heaps. Comparing this with the ground truth image (Fig. 6(a)) we see that the output is piecewise constant confirming the interchannel prior assumption. Further the residual image shown in Fig. 6(f) contains no structures and resembles that of added random noise.

Finally we show the result of L^1 norm M-estimator on 400×357 *Kid* color image taken from a mobile camera of 2 mega pixels resolution. This CCD sensor based image contains noise and blurring inherent to the imaging modality. We further add Gaussian noise $\sigma = 10$ to challenge the scheme. Proposed coupled RTV gives (Fig. 7(b)) sharp edges and selective smoothing in between as is evident from the background wall and the shirt. Simple region growing segmentation result is presented in Fig. 7(c) to show the versatility of the scheme. A profile line (Fig. 8) taken from the *Kid* image shows that our scheme has piecewise smooth behavior and color preservation. Table 1 lists the PSNR values of the images and quantitatively

(a) (b) (c)

Figure 7: Kid mobile camera image denoising: (a) original image (b) denoised using the proposed coupled L^1 norm scheme (3.3-3.4) (c) segmentation based on (b).



(a)

Figure 8: 1-D profiles: (dashed line) original and (solid line) denoised. Strong correlations between the channels are captured by our scheme without smearing color edges.

supports the observed visual improvements. PSNR of an estimate $\hat{u}$ of a true N-dimensional data u is defined by

$$PSNR(\hat{u}) = 10\ \log_{10}\left(\frac{N \times 255^2}{\sum_{x\in\Omega} |\Omega|^{-1}\,(\hat{u}(x) - u(x))^2}\right)\ dB$$

We also report the PSNR results of some well known benchmark color images from USC-SIPI database (images are not shown) in Table 2. This values quantitatively support the theory presented for the scheme (3.3) in this paper over other related schemes.

Note that in the coupled RTV scheme (3.3-3.4) only the regularization parameters λ, μ needs to be fixed for balancing the influence terms and the data fidelity term. One way to choose these

Table 1: PSNR (dB) values for various images

Images		$\sigma/PSNR$	Channel	L^1 norm	Lorentzian	KG	BKS
Dancer	(Fig. 3)	$20/13.44$	18.56	27.34	26.25	24.88	24.57
Monarch	(Fig. 4)	$30/19.56$	21.77	30.45	29.18	23.97	27.05
Spectral	(Fig. 5)	$15/8.87$	15.78	28.49	27.56	25.27	25.68
Colors	(Fig. 6)	$40/18.22$	22.20	34.75	33.87	30.89	31.13
Kid	(Fig. 7)	$10/30.89$	25.43	29.36	28.70	27.52	26.69

Table 2: PSNR (dB) values of standard test images under various schemes and for noise levels $\sigma = 10, 20$ and 30.

Ref.	σ	Lena	Pepper	Baboon	Barbara	F-16	Boat	Couple	House
	10	34.08	32.58	33.22	30.98	34.49	32.24	30.40	34.89
(2.5)	20	31.84	30.18	24.45	27.75	32.78	29.69	29.43	32.41
	30	29.69	29.00	22.77	24.42	29.69	26.85	28.13	30.25
	10	35.70	34.30	35.86	33.45	37.18	34.75	33.59	36.73
(3.3-3.4)	20	33.64	31.88	27.53	31.68	35.05	33.58	32.06	34.55
	30	32.21	30.97	25.74	28.74	32.47	30.33	31.93	32.40
	10	35.78	34.38	35.75	33.28	37.21	34.40	33.09	36.81
(3.3-3.5)	20	33.54	32.11	27.57	32.07	34.95	33.61	31.73	34.34
	30	32.09	30.64	25.20	29.55	32.30	30.12	32.11	32.27
	10	34.80	33.52	34.18	32.59	35.77	33.55	32.48	35.33
KG	20	32.02	30.59	25.94	31.72	33.68	32.24	31.72	32.90
	30	31.28	29.53	24.05	27.26	31.44	29.39	29.47	30.71
	10	34.72	33.34	34.25	32.43	36.08	33.71	32.71	35.28
BKS	20	32.78	30.68	25.43	31.12	33.81	32.69	31.68	33.13
	30	31.45	29.70	23.84	28.59	31.93	29.88	30.26	31.19

is to make them data adaptive, but in a multidimensional image data this is computationally expensive to do so. Recently, these hyperparameters were chosen using gamma prior distribution in super-resolution problems (Zibetti et al., 2008). It will be fruitful to extend the ideas to our restoration case. If we know a priori the background details of the imaging system then we can safely choose a parameter set for that particular instance. For example, after many experiments with RGB color images like Fig. 3 and Fig. 6 we found that λ in the range 10 to 20 and μ in 10 to 50 leads to similar results. Similarly for the coupled Lorentzian scheme (3.3-3.5) the parameter $k = 30$ works well for the mentioned real images.

Currently we are investigating other robust M-estimators and coupling function ψ for multispectral images. For example, Hampel's redescending function (Hampel et al., 1986) is non-convex yet it provides enhancement and better granularity control in gray scale images. Another non-convex M-estimator is the Tukey or Blake-Zisserman (Blake and Zisserman, 1987) function which gives zero response to outliers. On the other hand, among convex estimators L^1 norm is proven to be the best edge preserving prior in finite MRF setting (Ceccarelli, 2007) and our experiments confirm this assertion. A comparative study of other robust M-estimators combined with our coupled term for multidimensional data and to compute the regularization parameters in an adaptive way without increasing computational burden defines our future work.

5 Conclusion

Multichannel images have strong intrachannel correlations which are not taken care by channelwise restoration schemes. We proposed a novel MAP-MRF based model where apart from

the interchannel smoothing, intra channel edges are taken into account. This is achieved by applying an edge preserving M-estimator for all the channels of the input image separately and a coupling term which adds Gaussian prior based penalty for intra channel discontinuities. Weights are based on robust gradient estimations and can be made adaptive to handle high noise cases. Two M-estimators namely a convex L^1 norm and non-convex Lorentzian function were used and their results are compared on color and multispectral images. We proved the well-posedness of the corresponding energy minimization in the bounded variation space under some conditions on the interchannel prior. The scheme works for N-dimensional images and can be used as a preprocessing step in removing noise and outliers for multispectral images. The numerical results also show that the extra information provided by the coupling term stops the color smearing appearing and all channel edges are aligned in a coherent manner. Experiment with various images show the advantages of the proposed scheme.

References

Aubert, G. and Kornprobst, P. 2002. *Mathematical problems in Image Processing*, Springer-Verlag, New York.

Black, M. J., Sapiro, G., Marimont, D. and Heeger, D. 1998. Robust anisotropic diffusion, *IEEE Trans. Image Process.* **7**(3): 421–432.

Blake, A. and Zisserman, A. 1987. *Visual Reconstruction*, MIT Press.

Brook, A., Kimmel, R. and Sochen, N. A. 2003. Variational restoration and edge detection of color images, *J. Math. Imag. Vis.* **18**: 247–268.

Ceccarelli, M. 2007. A finite markov random field approach to fast edge-preserving image recovery, *Image and Vision Comput.* **25**: 794–802.

Geman, S. and Geman, D. 1984. Stochastic relaxation, gibbs distribution and the bayesian restoration of images, *IEEE Trans. Pattern Anal. Machine Intell.* **14**: 721–741.

Giusti, F. 1984. *Minimal surfaces and functions of bounded variation*, Vol. 80 of *Monographs Math.*, Birkhaüser, Basel.

Hampel, F., Ronchetti, E., Rouseeuw, P. and Stahel, W. 1986. *Robust statistics: the approach based on influence functions*, Wiley, New York.

Huber, P. J. 1981. *Robust statistics: the approach based on influence functions*, Wiley, New York.

Karen, D. and Gotlib, A. 1998. Denoising color images using regularization and correlation terms, *J. Vis. Comm. Img. Rep.* **9**(4): 352–365.

Li, S. Z. 1995. *Markov random field modelling in computer vision*, Springer, Berlin.

Li, S. Z. 1998. Map image restoration and segmentation by constrained optimization, *IEEE Trans. Image Process.* **7**(12): 1730–1735.

Rudin, L., Osher, S. and Fatemi, E. 1992. Total variation based noise removal algorithms, *Physica D* **60**: 259–268.

Szeliski, R., Zabih, R., Scharstein, D., Veksler, O., Kolmogorov, V., Agarwala, A., Tapen, M. and Rother, C. 2008. A comparative study of energy minimization methods for markov random fields with smoothness based priors, *IEEE Trans. Pattern Anal. Machine Intell.* **30**(6): 1068–1080.

Zibetti, M. V. W., Baźan, F. S. and Mayer, J. 2008. Determining the regularization parameters for super-resolution problems, *Signal Process.* **88**: 2890–2901.